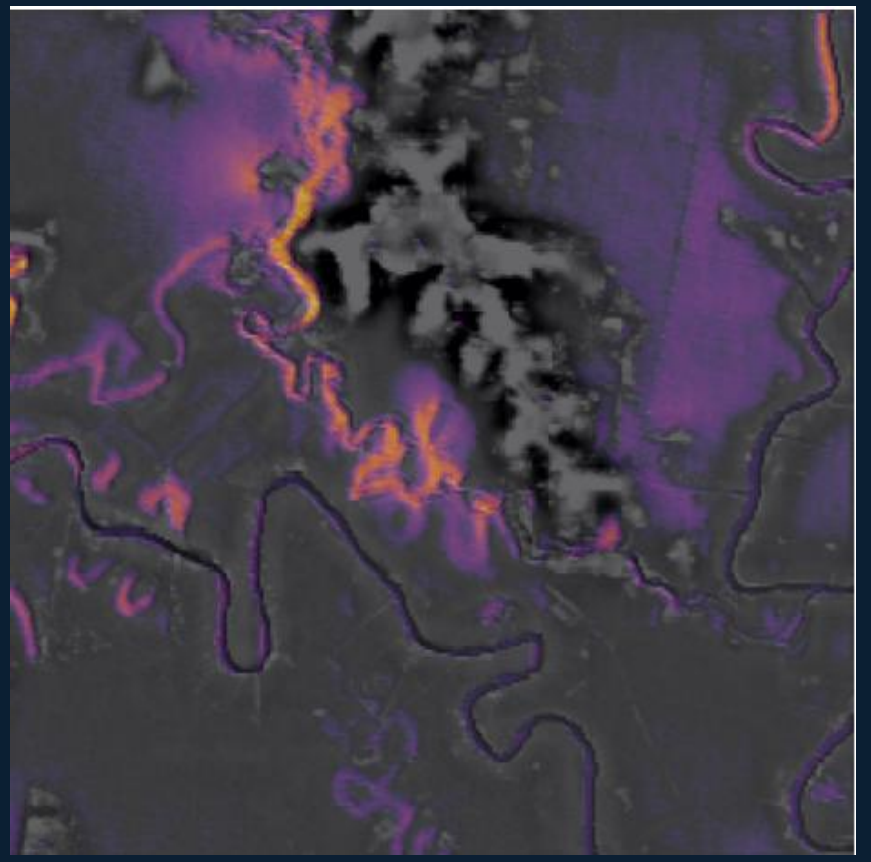
Deep Learning · CA3 Project

CNN-LSTM Based Spatiotemporal Flood Evolution Forecasting

Does a neural network forecast floods better than very simple rules?

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AIML A2 · October 2026

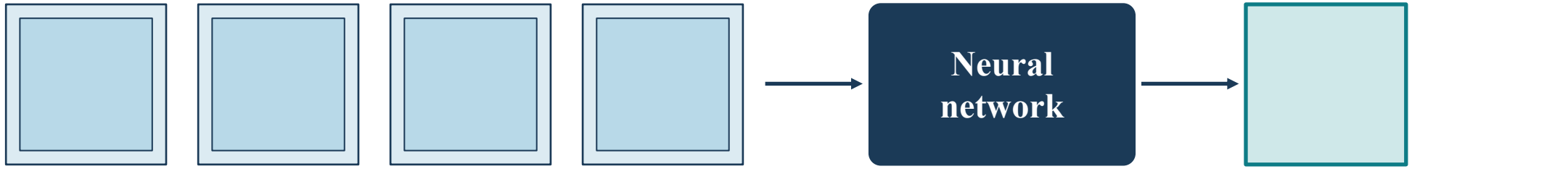


A picture from our model on the Australia flood. The bright lines are the river channels it pays most attention to.

The problem

Why a neural network, and what we ask it to do

Floods are fast and simulators are slow: one large flood can take days to simulate, but a trained network gives a forecast in seconds.



The last four depth maps, 30 minutes apart, plus the rain that has already fallen and the height of the ground.

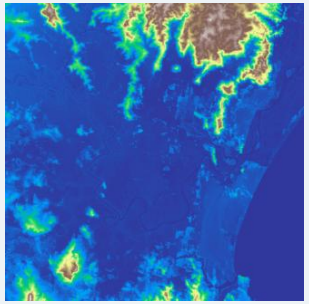
The catch

In 30 minutes the water in most places hardly changes. So just copying the last map is already a very good forecast. A neural network is only useful if it does better than simple forecasts like this one.

Part 1 (Colab notebook): one map, 30 minutes ahead. Part 2 (laptop study): five maps in a row, up to 2.5 hours ahead.

The data: four floods made by a simulator

Australia



20 Feb to 2 Mar 2022

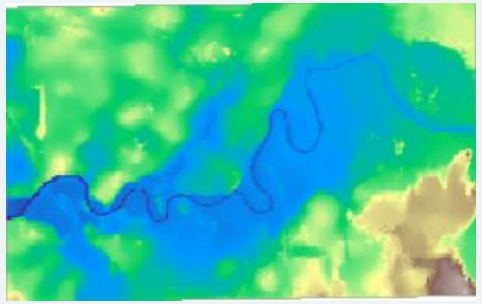
1073 × 1073 squares

each square 30 m wide

10 days, 481 maps

Used for training

UK



4 to 7 Dec 2015

170 × 275 squares

each square 30 m wide

3 days, 145 maps

Kept for testing

Pakistan



18 to 31 Aug 2022

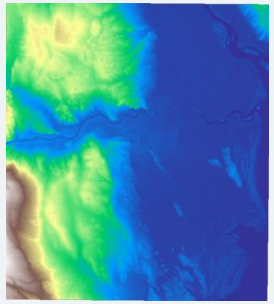
810 × 441 squares

each square 480 m wide

14 days, 673 maps

Used for training

Mozambique



14 to 20 Mar 2019

151 × 138 squares

each square 480 m wide

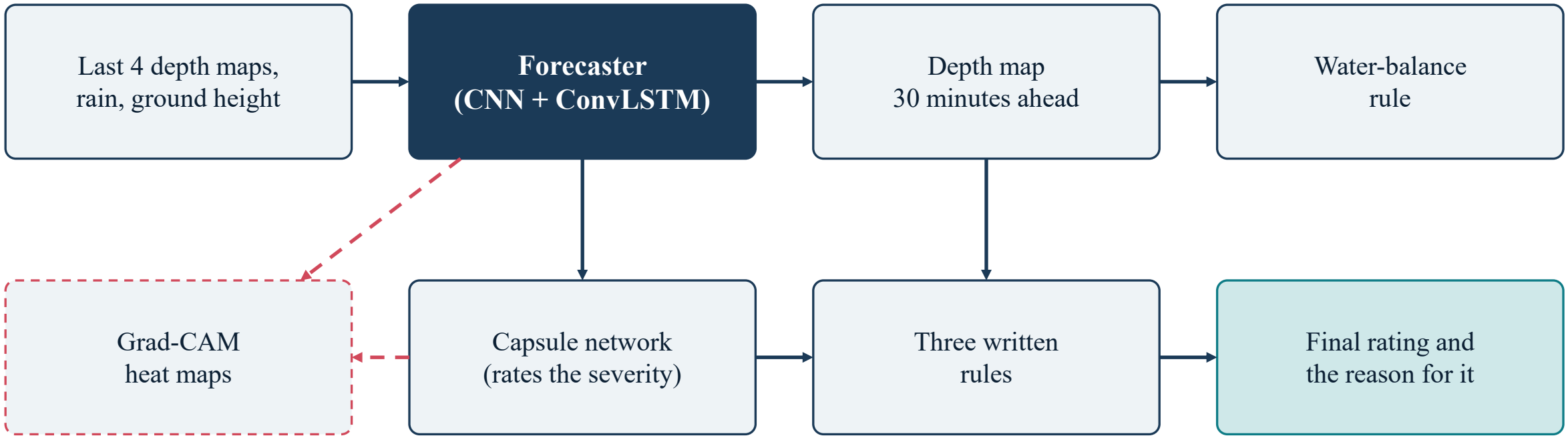
6 days, 289 maps

Kept for testing

We train on Australia and test on the UK (30 m squares), and train on Pakistan and test on Mozambique (480 m squares). Each training flood is split in time: the first 70% to train, the next 15% to decide when to stop, and the last 15% to test. The pictures show the ground height. All maps come from a physics simulator (FloodCastBench, Xu et al., 2025).

The system

Our whole system on one page



The forecaster predicts the next map, and the water-balance rule checks the total water against the rain. The capsule network rates how serious the flood is, and three written rules check that rating.

NeuroSymbolic AI: a neural network checked by rules people can read. The water-balance rule and the three written rules are our NeuroSymbolic part.

Explainable AI: showing why a model gave its answer. The dashed arrows are Grad-CAM, our way of doing that.

The model

The forecaster: how it reads the maps



- The CNN (convolutional neural network) layers look at each map on its own and pick out useful shapes, such as water edges, slopes and patches of rain.
- The ConvLSTM layer then reads the four maps in order and keeps a memory of how the water has been moving.
- The model predicts how much the depth will change, not the whole map. This is easier, because most of the map hardly changes in 30 minutes.

62,833

parameters, the numbers the model learns while it trains

How we score a forecast, and two versions of the model

skill = 1 − (error of the forecast ÷ error of copying the last map)

0 means the same as copying the last map, 1 means a perfect forecast, and below 0 means worse than copying.

Model A

starts from the last map

next depth = last depth + the change the model predicts
(and never below zero)

This was our first version. It beat copying the last map in three of the four tests, so it looked successful. But in the last part of the Australia flood its skill was −1.449, which is worse than copying.

Model B

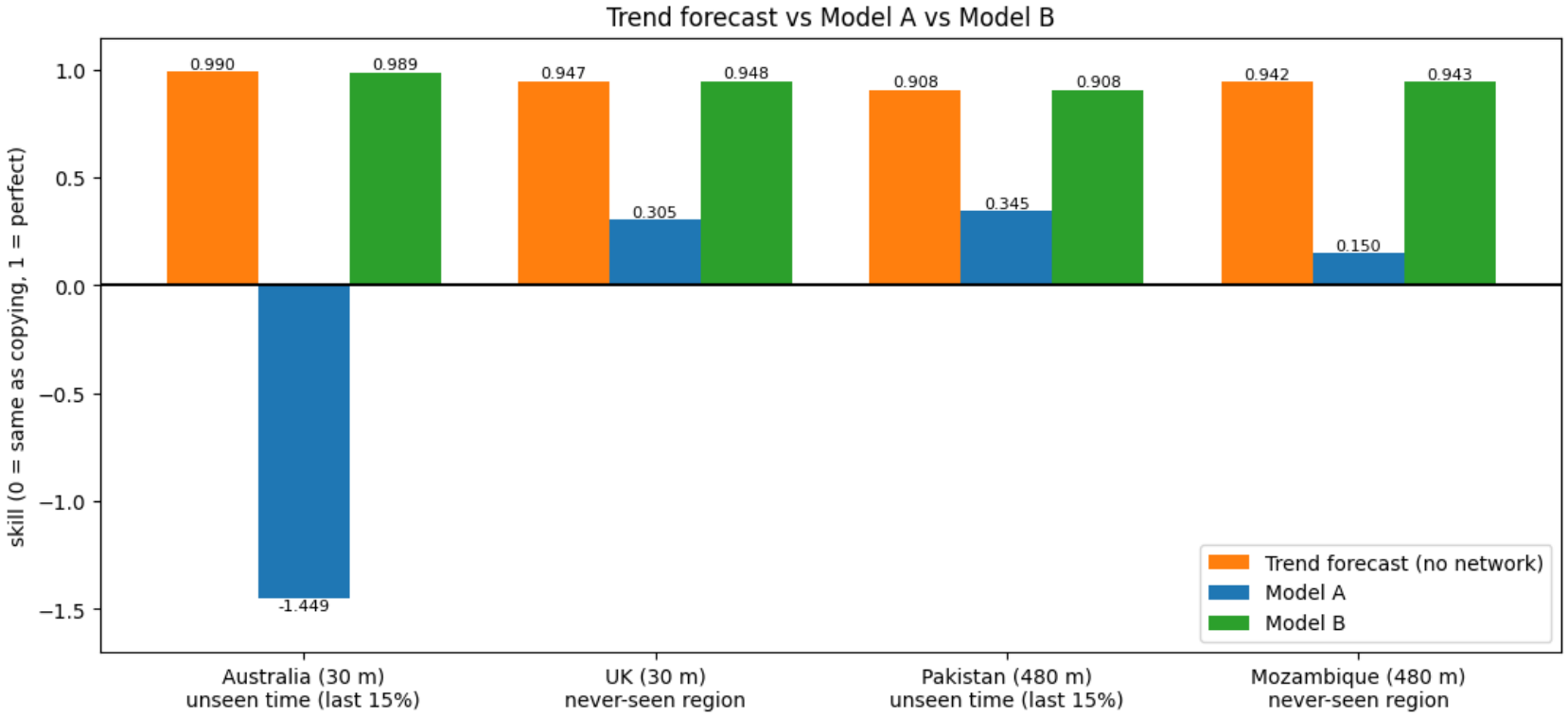
starts from the trend forecast

trend = last map + β × (last map − the map before it)
next depth = trend + a small correction from the model

The network only has to learn a small correction. β is one number found from the training data: 0.921 for the 30 m floods and 0.948 for the 480 m floods.

Part 1: the Colab notebook, forecasting 30 minutes ahead

The simple trend forecast beats our first model



−1.449

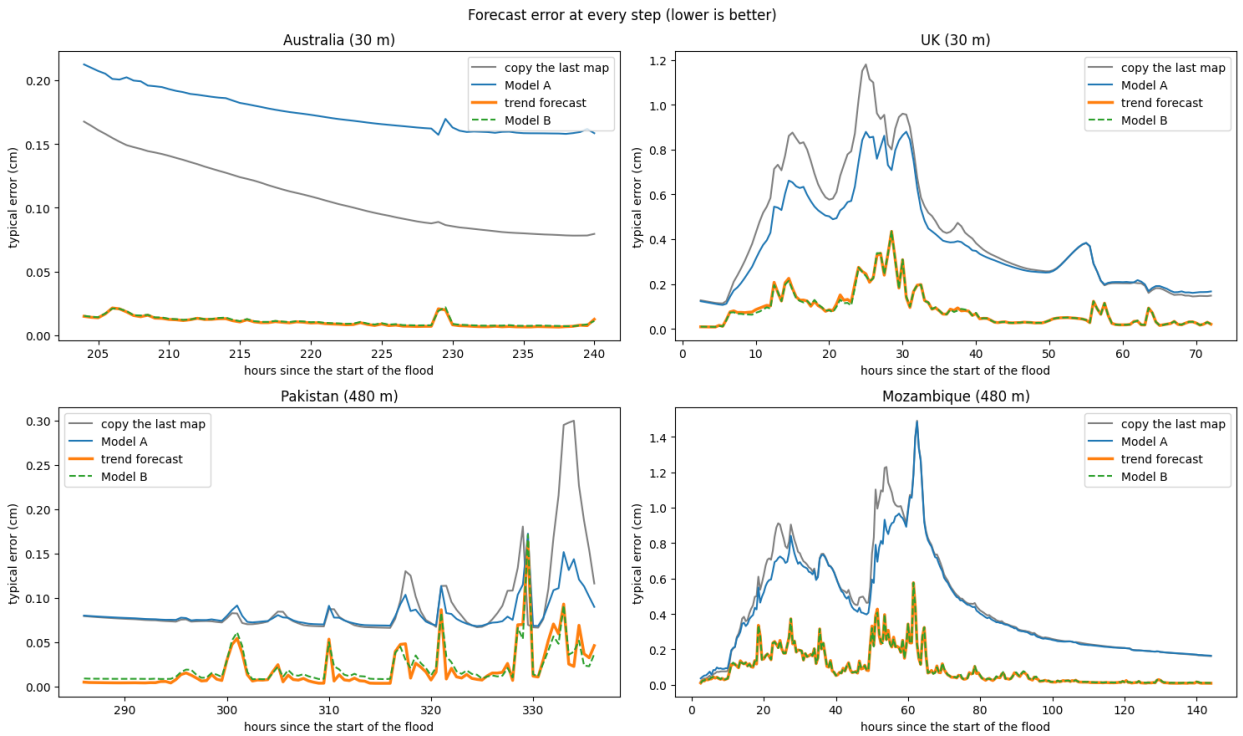
This is Model A in the last part of the Australia flood. It is worse than just copying the last map.

0.001

Model B stays within this of the trend forecast in every test. It matches the trend, but it does not beat it.

Higher is better. “Unseen time” is the last 15% of the flood the model learned from. “Never-seen region” is a different flood. All results are single runs.

At 30 minutes there is very little left to improve



1%
Australia

5%
UK

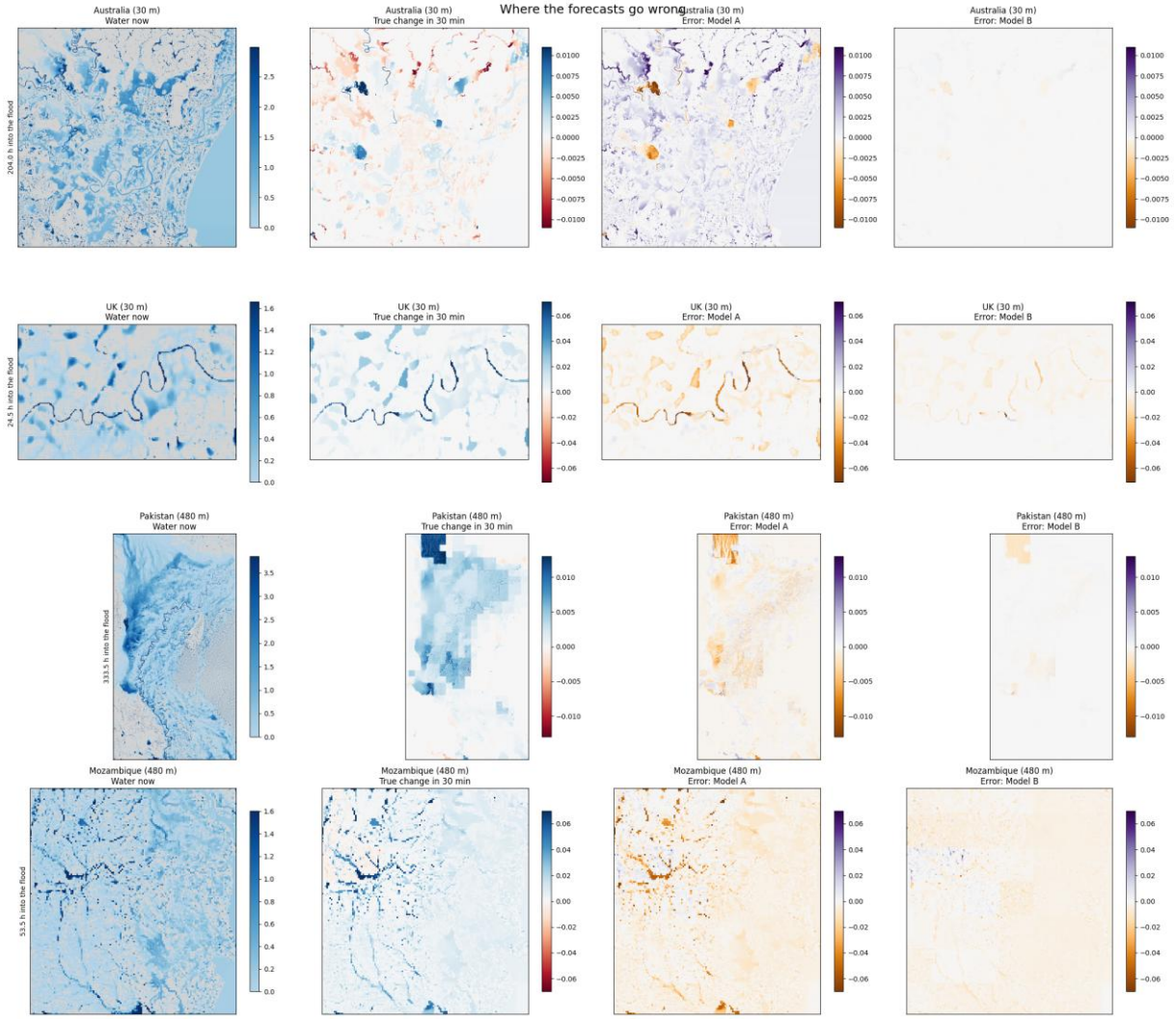
9%
Pakistan

6%
Mozambique

How much of the error of copying the last map is still left after the trend forecast. The chart shows the typical error over time for each flood.

So Model B matches the trend because almost nothing is left to add. A fair test of the LSTM's memory needs longer forecasts, 6 to 24 hours ahead.

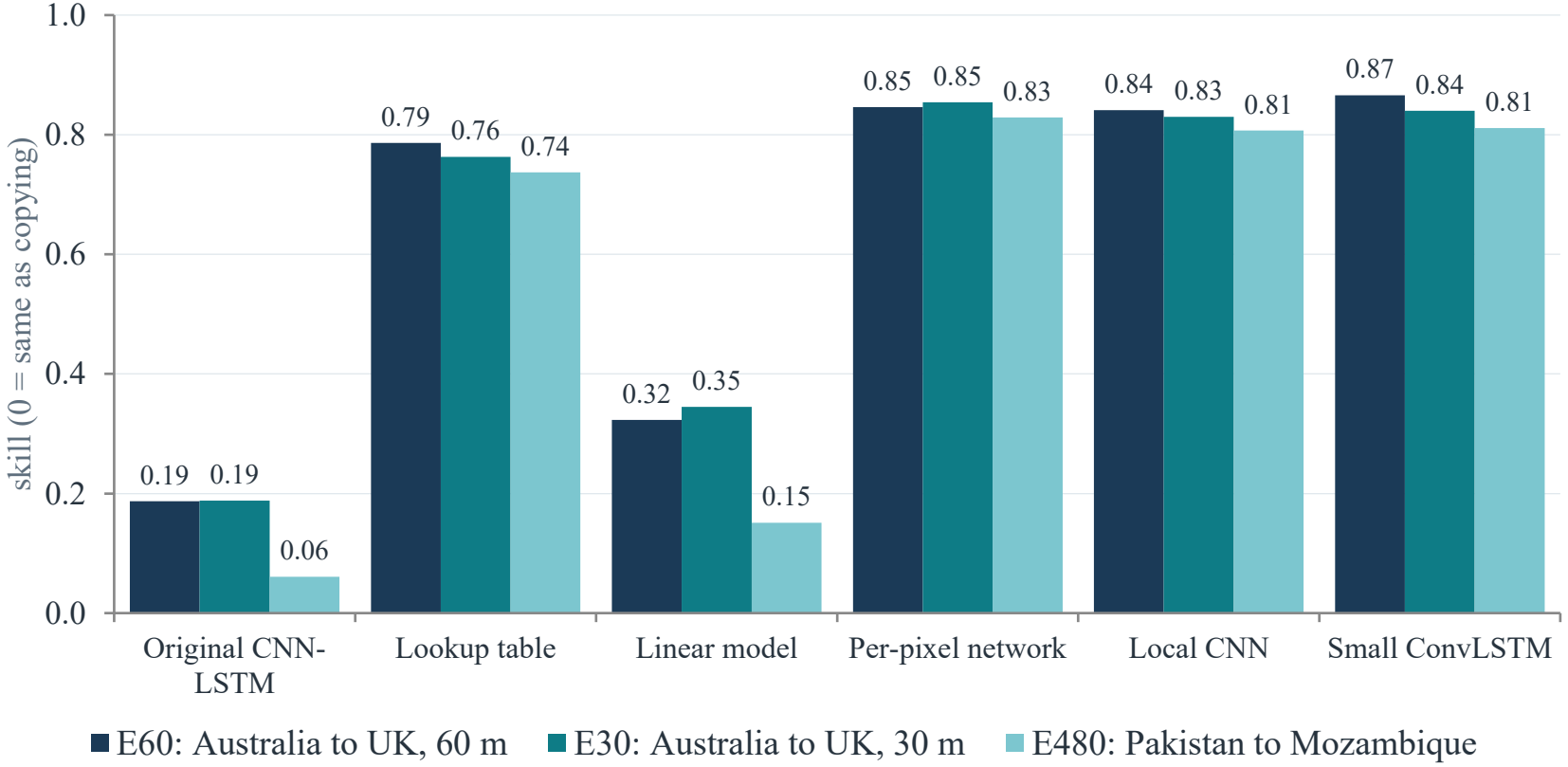
On the map, Model B's mistakes almost disappear



One row per flood: Australia and the UK (30 m squares), then Pakistan and Mozambique (480 m squares). Purple means too much water was forecast, and orange means too little. The last column, Model B's error, is almost blank.

Part 2: a bigger study on a laptop, forecasting up to 2.5 hours ahead

A simple lookup table already gets most of the way



0.74 to 0.79

is the skill of a lookup table that only uses the current depth and the last change of each square.

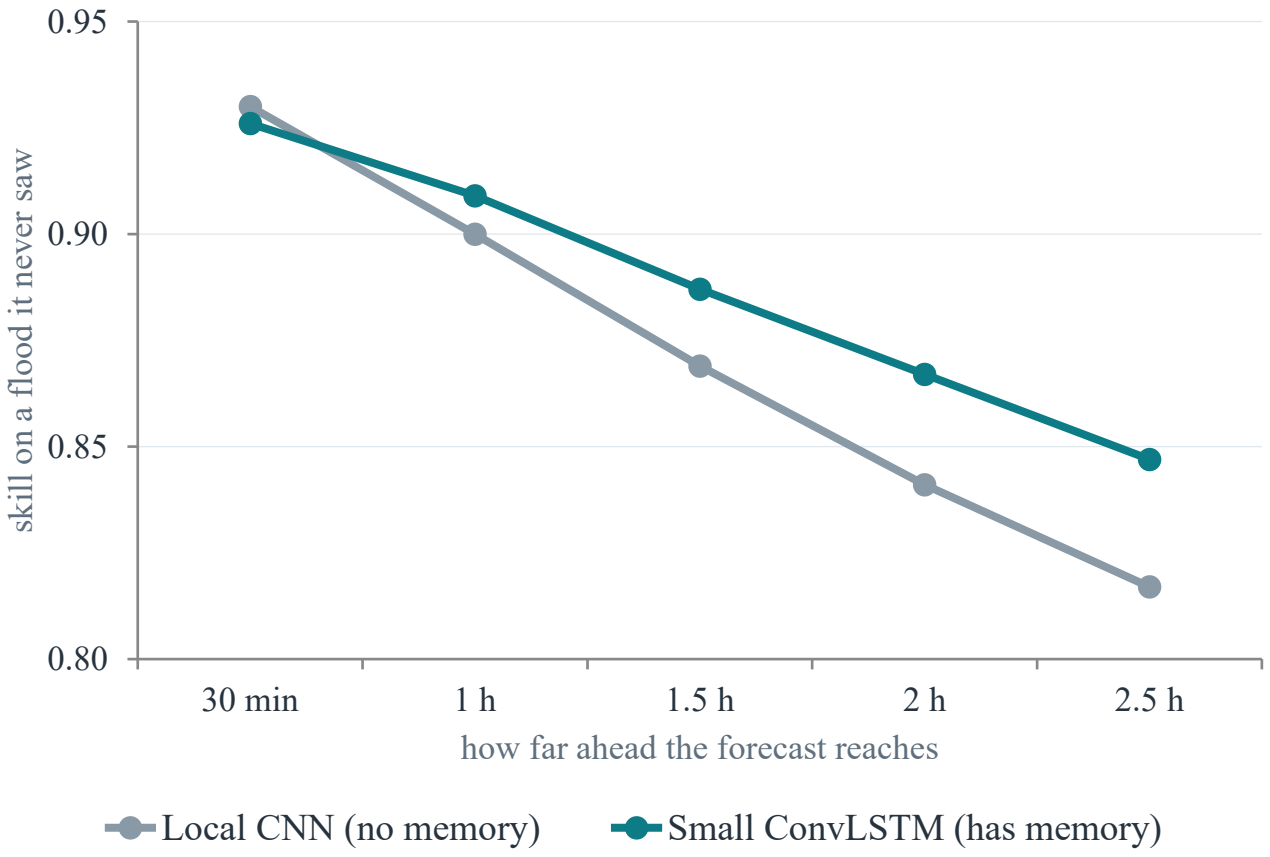
+0.05 to +0.08

is all that the local CNN and the small ConvLSTM add on top of it.

About 310 training runs. E60 and E30: trained on Australia, tested on the UK at 60 m and 30 m. E480: trained on Pakistan, tested on Mozambique. The linear model scored low because the way we fitted it gave almost no weight to how the depth was changing. Per-pixel network: a small network that works on one square at a time.

Part 2: the laptop study

Does the LSTM's memory help?



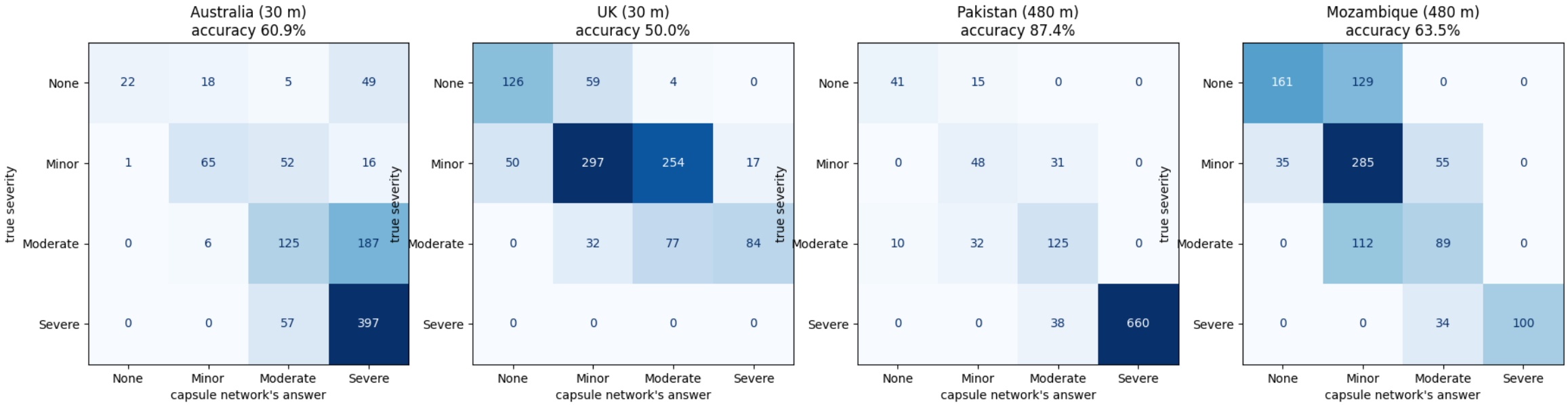
When both models are trained for the same length (90 passes through the training data), they score the same. The difference is +0.004 on the UK and −0.014 on Mozambique, inside the ±0.03 band we set in advance to mean “the same”. A model 140 times bigger did not do better either.

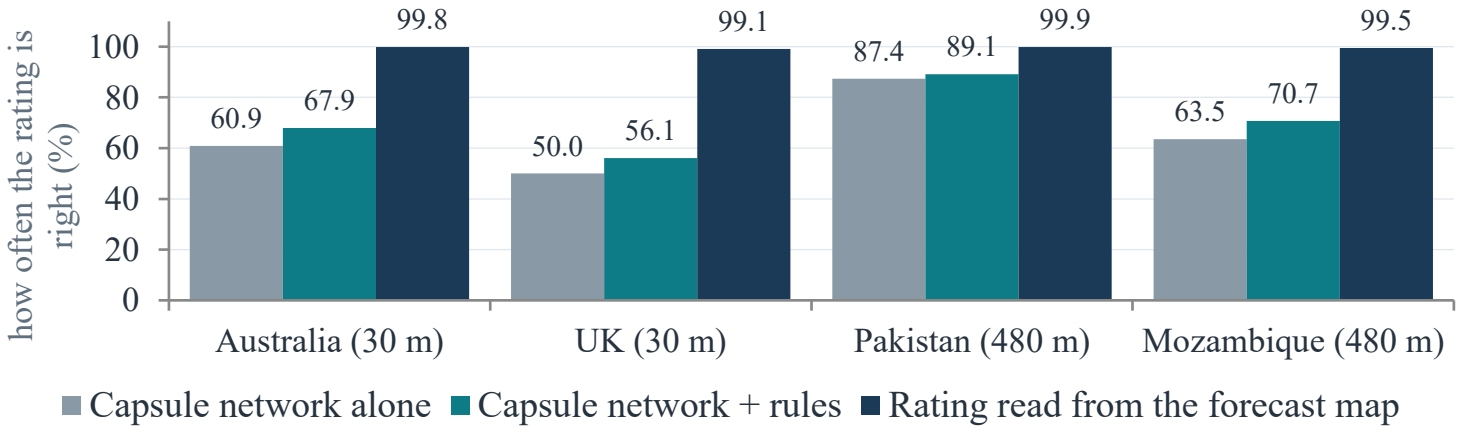
The ConvLSTM does lose less skill as the forecast reaches further ahead. At 30 minutes the plain CNN is slightly better, and at 2.5 hours the ConvLSTM is better. It also stayed stable when we hid the start of a flood from training.

We did not test whether the memory is the reason for these two differences.

The chart shows Australia to UK at 60 m, with the planned 30-pass training. The other two experiments show the same pattern.

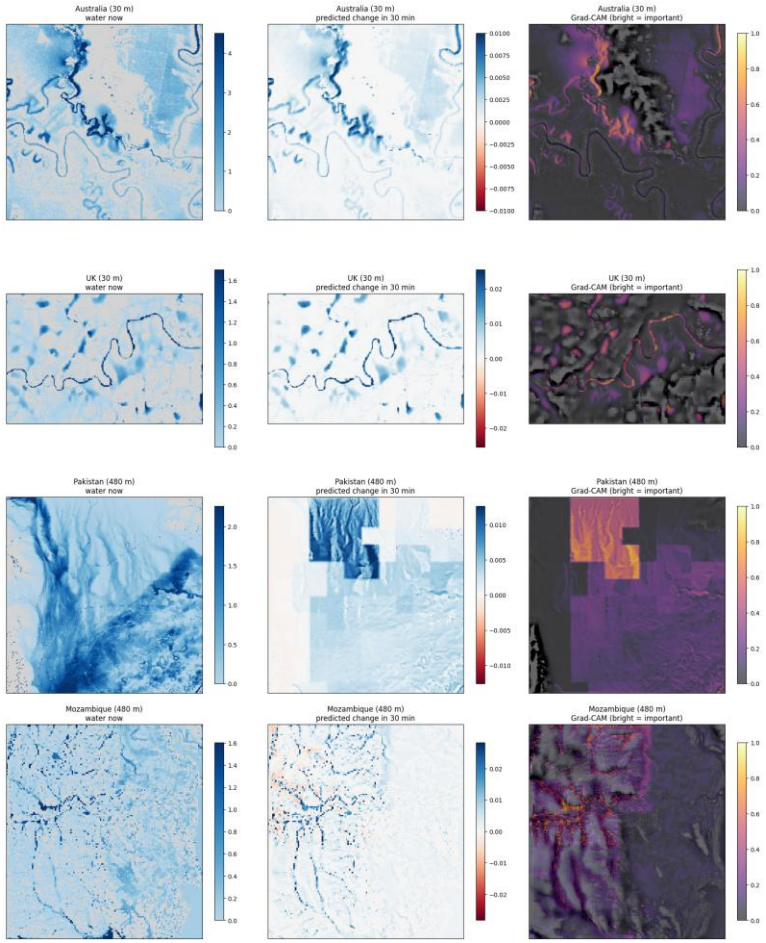
Rating how serious the flood is



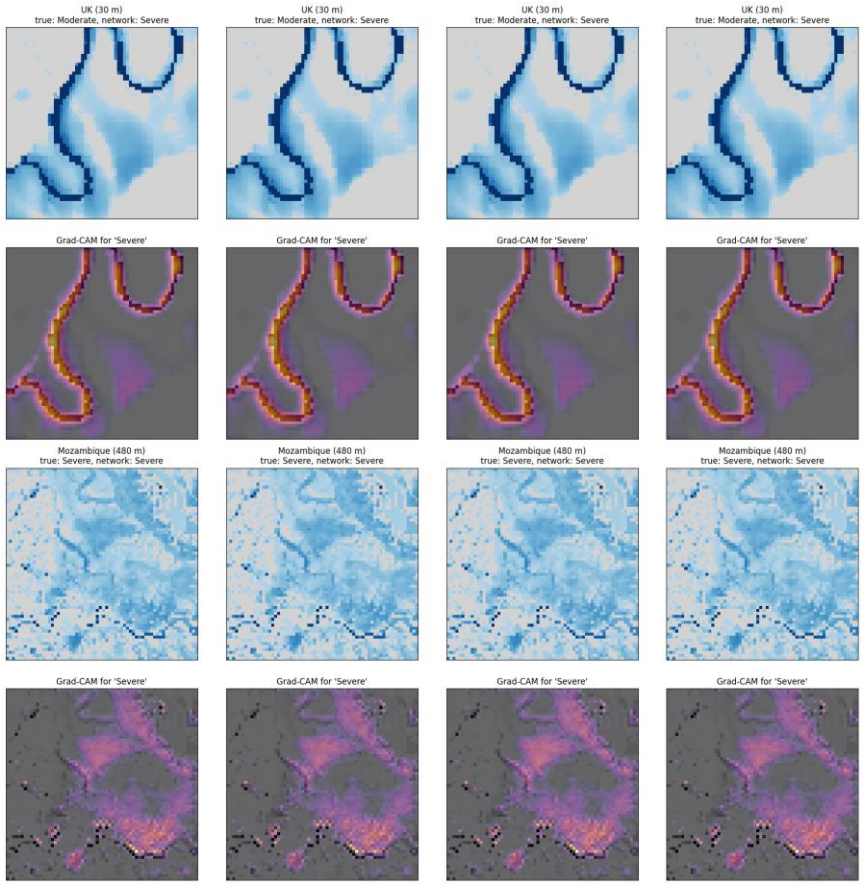


- Three written rules check the capsule network's rating and always give a reason. They raised the accuracy in all four floods.
- The rating read straight from the forecast map is right about 99% of the time, because the rating in 30 minutes is almost the same as now.

Where do the models look?



The forecaster: water now, predicted change, Grad-CAM



The capsule network on the UK and Mozambique

Explainable AI means showing why a model gave its answer. Grad-CAM colours the parts of the input that mattered most, and bright means more important.

For the forecaster

The bright areas follow the river channels and flood edges, where the water is changing.

For the capsule network

The heat maps were weak. Two training runs agreed only a little, so we show them as pictures, not proof.

Conclusion

What we learned

1. At 30 minutes ahead, a simple trend forecast with one fitted number does most of the work. Our second model matches it, but it does not beat it.

2. Up to 2.5 hours ahead, a simple lookup table that only uses each square's current depth and last change already scores 0.74 to 0.79. The best networks score 0.81 to 0.87, and the LSTM's memory did not help when the training length was equal.

3. Always compare a model with simple forecasts. Our first model looked good until we did.

Limits: we have only one flood per region, so only two truly independent train-and-test pairs. The floods are simulated, not real. The Colab results come from single runs.

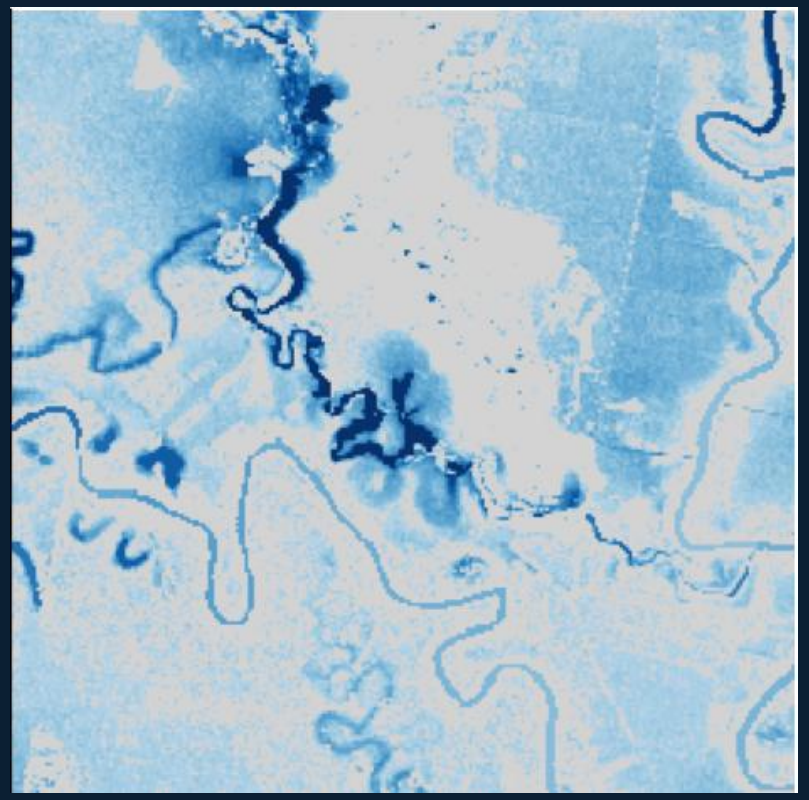
What we want to do next: forecast 6 to 24 hours ahead, where a simple trend should get worse and the LSTM's memory could really matter. We also want to rebuild the capsule network and Grad-CAM on Model B, and repeat the Colab runs with several random seeds.

Thank you

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Main references

Q. Xu, Y. Shi, J. Zhao and X. X. Zhu, FloodCastBench, Scientific Data, 2025.
X. Shi et al., Convolutional LSTM network, NIPS 2015.
S. Sabour, N. Frosst and G. E. Hinton, Dynamic routing between capsules, NIPS 2017.
R. R. Selvaraju et al., Grad-CAM, ICCV 2017.
The full list of 22 references is in the report.



The water depth in the Australia flood, one step before the forecast.